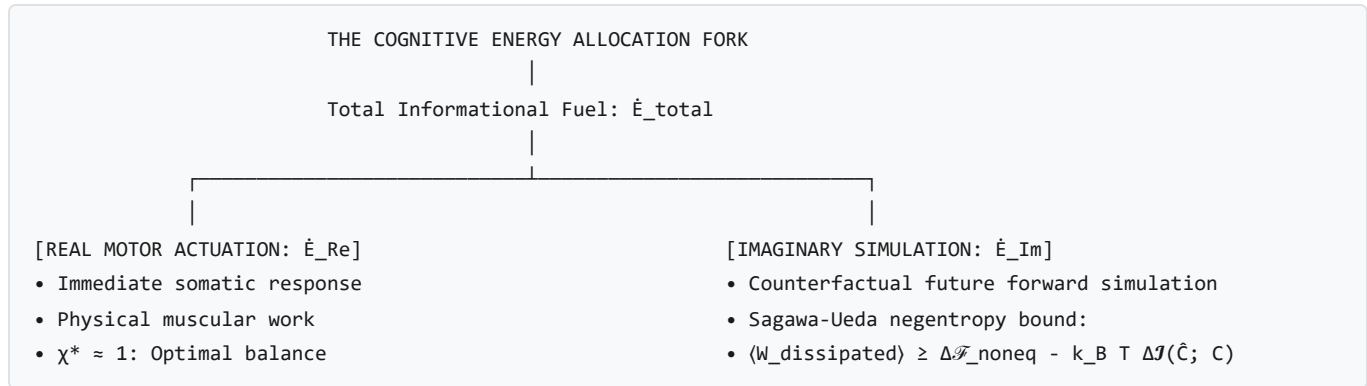


Cognitive Thermodynamics, The Ariṣadvarga, and the Landauer Veto

Cognitive & Psychological Focus: This treatise formalizes **Tier III Cognitive Architectures**, **Bayesian Active Inference Engines**, the **Thermodynamics of the Six Internal Afflictions (Ariṣadvarga)**, and the **Landauer Information-Erasure Cost of Voluntary Veto (Viveka)**.

1. Tier III: Cognitive Agents as Complex Predictive Engines

In [draft.md §2.3.4 & §4](#), a cognitive agent ($E \in T_{III}$) operates across complexified phase space $\Omega_{\mathbb{C}} = \Omega_{\mathbb{R}} \oplus i\Omega_{\mathbb{I}}$: * **Real Substrate** ($\Omega_{\mathbb{R}}$): Immediate somatic, motor, and neurochemical configurations. * **Imaginary Substrate** ($\Omega_{\mathbb{I}}$): Internal generative Bayesian models simulating counterfactual future trajectories ($\hat{\mathbf{C}}_{\text{future}}$).



The Sagawa-Ueda Information-Thermodynamic Bound:

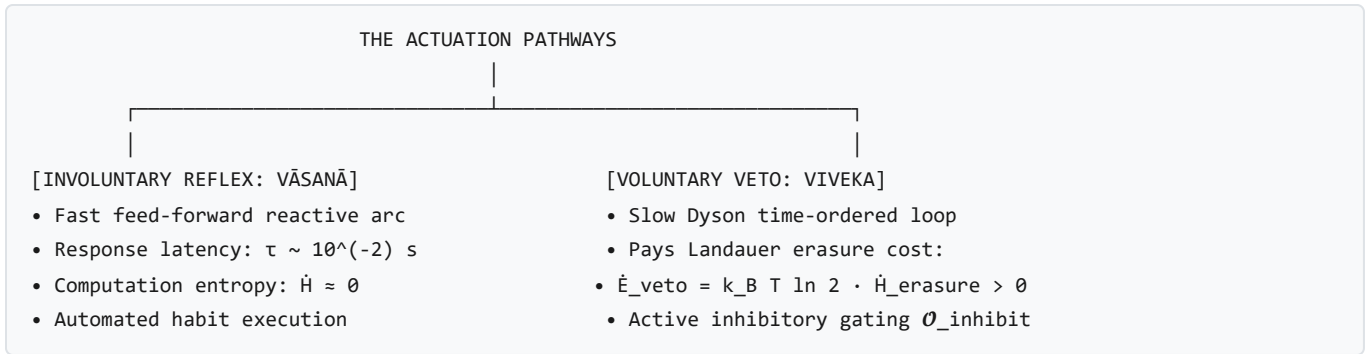
The dissipation required to survive an environmental shock is reduced by the mutual information $\Delta \mathcal{I}(\hat{\mathbf{C}}; \mathbf{C}_{\text{future}})$ acquired via internal simulation:

$$\langle W_{\text{dissipated}} \rangle \geq \Delta \mathcal{F}_{\text{noneq}} - k_B T \cdot \Delta \mathcal{I}(\hat{\mathbf{C}}_{\text{predicted}}; \mathbf{C}_{\text{future}})$$

Pre-stiffening cognitive priors ($\mathbf{R}_{\text{active}}(t - \Delta t)$) reduces physical damage to near zero upon predictable impacts.

2. Involuntary Latency (*Vāsanā*) vs. Voluntary Landauer Veto Gating (*Viveka*)

Living cognitive agents possess two distinct modes of interfacial actuation:



• **Involuntary Reflex (Vāsanā / Saṃskāra):**

- An automated feed-forward reflex arc where incoming challenge traction $\mathbf{C}$ directly triggers motor expression $\mathcal{X}$ through pre-configured neural pathways ($\tau_{\text{latency}} \sim 10^{-2}$ s). Because no deliberative computation or state re-evaluation occurs, the Landauer informational entropy cost is zero ($\dot{H} \approx 0$).

• **Voluntary Discriminative Veto (Viveka):**

- The conscious intervention of the higher-order Dyson loop ($\mathcal{T} \exp(\int \hat{\mathcal{L}} d\tau)$), which inspects the automated trajectory and applies an active inhibitory gate ($\mathcal{O}_{\text{inhibit}}$) to prevent expression.
- **The Landauer Metabolic Tax:** Erasing the automated motor command requires physical bit erasure, demanding continuous free-energy dissipation:

$$\dot{\mathcal{E}}_{\text{veto}} = k_B T \ln 2 \cdot \dot{\mathcal{H}}_{\text{erasure}} > 0$$

Exercising conscious restraint (Viveka / Dama)* is physically exhausting because it consumes real biochemical glucose ($k_B T \ln 2$) to clear neural register buffers.

3. Top-Down Craving (Kāma) vs. Bottom-Up Hunger (Kṣudhā)

Hunger (Kṣudhā*): A bottom-up First-Law exergy deficit occurring entirely in real configuration space $\Omega_{\mathbb{R}}$:

$$\dot{E}_{\text{fuel}} < \dot{E}_{\text{crit}} \equiv T_{\text{ambient}} \int_E \sigma_{\text{total}} dV \implies \frac{d\mathcal{G}}{dt} < 0$$

Kṣudhā is a biochemical indicator of metabolic depletion (falling blood glucose, low ATP/AMP ratio) requiring zero memory simulation.

Desire / Craving (Kāma / Trṣṇā*): A top-down simulation executed in imaginary phase space $\Omega_{\mathfrak{Jm}}$:

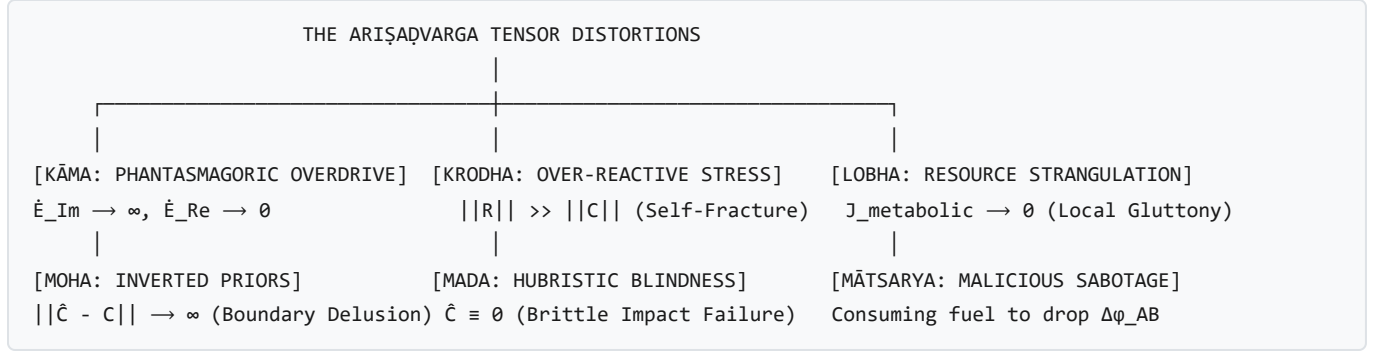
$$D_{\mathfrak{Jm}} = \hat{\pi}(\mathcal{F}_{\text{ledger}}) \xrightarrow{\text{simulation}} \Delta \hat{\mathcal{G}}_{\text{predicted}} > 0$$

Kāma occurs when the cognitive model recalls past reward states from $\mathcal{F}_{\text{ledger}}$ and projects an artificial free-energy surplus. This top-down prediction transduces downward into physical physiology via the Semantic Transduction Tensor (

$\mathbf{C}_{\text{craving}} = \mathbf{K}_{\text{trans}} \cdot \nabla_{\theta} \mathcal{D}_{\text{KL}}$), generating visceral dopamine cascades and restlessness **even when fully satiated**.

4. Thermodynamics of the *Ariṣaḍvarga* (The Six Internal Afflictions)

In classical Sanskrit psychology, the *Ariṣaḍvarga* represents the six inner enemies that destroy discernment. In our framework, these are derived as **exact mathematical distortions of the structural margin tensor** $\phi(x, t)$:



- **Kāma (काम — Unbounded Imaginary Projection Overdrive):**
 - Informational fuel allocation diverges ($\chi \rightarrow \infty \implies \dot{\mathcal{E}}_{\text{Im}} \rightarrow \dot{\mathcal{E}}_{\text{total}}, \dot{\mathcal{E}}_{\text{Re}} \rightarrow 0$). The entity expends all free energy simulating ungrounded imaginary prospects while the physical substrate starves into anergic boundary collapse.
- **Krodha (क्रोध — Explosive Over-Reactive Stress Discharge):**
 - Internal counter-stress vastly exceeds external challenge ($\|\mathbf{R}_{\text{active}}\| \gg \|\mathbf{C}\|$). The localized stress concentration generates internal Rankine-Hugoniot shock waves that fracture the entity's own boundary from within.
- **Lobha (लोभ — Parasitic Resource Hoarding):**
 - A constituent node hoards fuel measure ($\mu(E^j) \uparrow$) while choking metabolic syncytial flux ($\mathbf{J}_{\text{metabolic}}^{j \rightarrow k} \rightarrow 0$). This starves neighboring nodes, triggering a localized ischemia cascade that collapses the macro-envelope.
- **Moha (मोह — Perceptual Delusion / Model Inversion):**
 - The generative model divergence explodes ($\|\hat{\mathbf{C}} - \mathbf{C}_{\text{true}}\| \rightarrow \infty$). The entity mistakes destructive threats for fuel sources, actively opening its boundary pores ($\mathbf{v}_n \cdot \hat{\mathbf{n}} < 0$) to toxic foreign challenge fields.
- **Mada (मद — Hubristic Invulnerability / Zero-Challenge Prior):**
 - The entity sets its predicted external challenge to zero ($\hat{\mathbf{C}} \equiv 0$), disarming its adaptive viscoelastic dampers ($\nu \rightarrow 0$). Upon the arrival of real environmental shocks, it undergoes catastrophic brittle fracture without damping.
- **Mātsarya (मात्सर्य — Malicious Zero-Sum Trophic Sabotage):**
 - An entity expends its own metabolic exergy solely to diminish a neighbor's margin ($\Delta\phi_{\text{AB}} < 0$), lowering total syncytial free energy and precipitating mutual systemic lysis.